



APOGEE

QAU SCIENCE SOCIETY MAGAZINE

HORRORS OF SCIENCE

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Apogee Magazine is the official yearly publication from Quaid-i-Azam University Science Society including articles/reviews/opinions in Science and Technology along with faculty/researcher interview(s) and society events.

“Special thanks to Dr. Muhammad Jamil Aslam, professor at Department of Physics, Quaid-i-Azam University, for making the students’ ambitions a reality and for being a constant beacon of support and guidance.”

APOGEE

QAU SCIENCE SOCIETY MAGAZINE

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Horrors of the science

Science is a paradox

By Usman Suhail

A dazzling creator of wonders and a chilling instrument of horrors! This duality shines in its triumphs: unveiling the universe's hidden mysteries, conquering once-incurable diseases, and propelling humanity to the galaxy's distant edges. Yet, it has also unleashed nightmares, inflicted untold pain, and devoured countless lives. The fault lies not in scientific knowledge itself, but in how it's wielded and twisted.

The starkest, most haunting proof of this split nature emerges from the atrocities committed under the banner of science during the major wars in history. The Nazi concentration camps stand as a grim testament: when science is stripped of wisdom, compassion, and ethics, it morphs into a weapon of slaughter and ruin. Earlier, the Second Reich's blind faith in evolutionary theory and social Darwinism fueled the 20th century's first genocide, the systematic extermination of South Africa's indigenous peoples. These examples scream a brutal truth: science, unchecked, can turn monstrous.

In warfare, science's destructive streak roars even louder. Physicist Enrico Fermi, a Nobel laureate, captured it perfectly: **"We knew the world would not be the same."** The atomic bomb, a breathtaking feat of physics, ushered in an age of unthinkable devastation. Hiroshima and Nagasaki weren't just military moves but they were thunderous warnings of science's apocalyptic potential when bent toward annihilation. Japanese novelist Kenzaburō Ōe, who grew up in their shadow, put it piercingly: **"Hiroshima and Nagasaki were the moment when mankind lost its innocence... a terrifying, terrible moment that defines what humanity is capable of."**

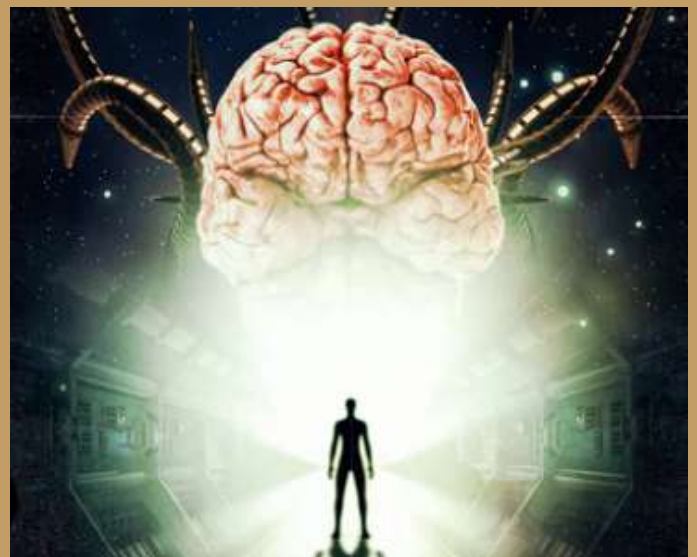
Now, as we teeter on the brink of a biotech revolution with genetic engineering in our grasp, new ethical storms loom. Altering an organism's genetic code stirs profound questions about identity, morality, and the boundaries of human meddling. The quest for a **"genetically perfect human"** flirts with a modern eugenics, threatening equality, diversity, and the very soul of what it means to be human.

A deep dive into science's past lays bare a sobering reality: knowledge without morality rivals ignorance in its peril. Science isn't evil by nature; its power simply demands accountability. By facing its shadowy side head-on, we can steer

it toward a brighter future for all.

**"Our world is wonderful and full of possibilities,
but it's also full of very dark possibilities,
and they are all tied to the misuse of science."**

— Carl Sagan



Nobel prize in medicine and physiology 2024



Nobel prize in medicine and physiology 2024

By : Zainab

The Nobel Prize in Medicine or Physiology 2024 was awarded to two brilliant minds: Victor Ambros of University of Massachusetts Medical School, and Gary Ruvkun of Harvard Medical School and Massachusetts General Hospital. Their work led to the discovery and mechanism of working of micro-RNA (miRNA), a minuscule molecule involved in post-transcriptional gene regulation.

To understand how this finding came to be and what it entails, basic background knowledge needs to be instilled. Every organism is built and maintained by a molecule known as DNA. It contains genetic information, which 'codes' for or makes different proteins that constitute the structures of a body. It does this by first making an mRNA molecule in a process called transcription. This acts as a sort of mobile messenger between the DNA stage and the protein stage. Most RNAs are 'coding,' that is, they make proteins through translation. Transcription and translation are highly regulated through complex strategies in different stages. To summarise, information is passed from DNA to RNA to proteins in a controlled series of procedures. In the 1990s, Ambros and Ruvkun were working with a species of nematode called *C. elegans*. The study of development-related genes of the organism brought to their attention a small molecule, later termed the miRNA. This RNA molecule, which was produced by a gene involved in development, was not coding.

Rather, it was involved in the aforementioned regulatory methods. It was found bound to an mRNA molecule due to which translation was inhibited, effectively stopping the formation of associated proteins. Later in the year 2000, Ruvkun's team discovered another miRNA molecule that was present not only in *C. elegans* but also in other species, including humans, demonstrating that miRNAs are not species-specific regulatory techniques.

This breakthrough had numerous implications for human health. Since the miRNA molecules regulate critical cellular processes, their dysregulation or improper working can lead to diseases and disorders, many of which have been identified. They have been shown to cause cancer through increased expression, cardiovascular diseases through initiating inflammatory responses by the immune system, and neurological disorders such as neurodegenerative diseases. As such, miRNA have been utilised for therapeutic endeavours. They can be targeted and their expression rates controlled, or even 'edited' or modified to correct any mistakes in the structure of the molecule.

The work of Ambros and Ruvkun illustrates the vital nature of molecular biology and such fruitful innovations in the improvement of human health and life expectancy. They not only uncovered a major biological mechanism, but their contribution has also been harnessed to achieve new heights. The prestigious Nobel Prize awarded to them underscores this.

Microbiome and Mood Disorders

How gut microbiota shapes our mental well-being

BY: Bisma Khan

Have you ever wondered if microbes, which are some of the tiniest creatures on this planet, can affect our mental health? Yes, these tiny creatures, which are not even large enough to be seen by the naked eye, can have monumental visible effects on our mental behavior and psychological well-being. Any irregularity in the composition of the microbiome residing in our body is responsible for many mood disorders like depression and anxiety.

Human Microbiome

The microbiome is the group of all microorganisms residing in our body. Surprisingly, we have more microbes in our body than body cells. According to PubMed Central (PMC), our body is composed of approximately 30 trillion cells and 38 trillion bacteria.

These trillions of micro-organisms are not merely residing in our bodies but hugely influencing our moods and mental health as well.

History

The discovery of microbes in humans goes back to 19th century. In 1822, a North American trader Alexis St Martin was shot in lower abdomen which severely injured his stomach. Fortunately, his life was saved by an army surgeon William Beaumont through multiple surgeries. However, he couldn't treat his stomach's hole. He conducted a series of experiments to examine its effects on the behaviour of St Martin. He discovered a shocking relation between human gut and its effect on the patient's moods and emotions. Due to the discovery of this gut-brain relation, Beaumont was called the Father of Gastric Physiology.

How is our mental health connected with microbiota?

Our gut contains the majority of human microbes and is linked with the brain via the gut-brain axis. It is a bidirectional pathway that

covers the immune system, metabolic system, gastrointestinal system, and endocrine system. Anatomically, our gut microbiota is connected with the brain through the vagus nerve. This indicates that our brain is in constant communication with the gut microbiota, reacting in response to signals from gut microbes. Our gut is also termed the second brain of the body because of its ability to produce neurotransmitters and stimulate dopamine and serotonin, which are directly linked with our mood, thoughts, behaviors, and cognition.

Microbiome and Mood Disorders

According to a study by the National Library of Medicine, mental disorders like anxiety and depression are related to gut microbiota imbalance (dysbiosis). Another research study by Biological Psychiatry revealed that psychiatric conditions like bipolar disorder, insomnia, and depression are also linked with the microbiome of our gut.

Shaping mental health via microbiota:

Well, now that we are quite aware of the role of microbiota in shaping our mood through hormonal balance, we should take care of these tiny entities for improved mental health. We can maintain the balanced composition of our microbiota by having a simple and healthy diet. Including prebiotics, probiotics, and dietary fiber in our meals will help us grow a healthy and beneficial microbiome. It will not only improve our microbiome but also modulate the structural and psychological changes in our brain. Also, improving our lifestyle by adding physical activities and actions will aid us in stress management by supporting our microbiome and emotional well-being. To put it in a nutshell, microscopic microbes have a mighty effect on our mental health. So, take care of these tiny creatures if you want to unlock the secrets of happy life.

Microflora and it's role in Health and well being

Your Body's Tiny but Mighty Allies

By: Warda Tahir



Inside your body, trillions of microscopic organisms—bacteria, fungi, and viruses—are hard at work. This invisible ecosystem, called the microbiome, plays a huge role in keeping you healthy. From digestion and immunity to mental well-being and aging, these tiny life forms influence it all. When your microbiome is balanced, you thrive. But when it's thrown off—something called dysbiosis—it can lead to various health problems. Scientists are now uncovering how we can support and harness this microscopic world to improve health and even prevent diseases.

Your Gut and Your Bones—A Surprising Connection

Did you know that the health of your gut bacteria affects your bones? Research shows that an imbalanced gut microbiome is linked to conditions like osteoporosis, which weakens bones and increases the risk of fractures. Your gut bacteria help regulate calcium absorption, bone metabolism, and inflammation—all crucial for strong bones. The good news? Eating foods rich in probiotics, prebiotics, and fiber can help keep both your gut and bones in top shape.

Can Your Microbiome Slow Down Aging?

As you age, your microbiome changes, and not always for the better. Some beneficial bacteria, like *Akkermansia* and *Butyrivibrio*, are linked to longer lifespans and better overall health. Scientists are exploring ways to boost these microbes through diet, regular exercise, and even microbiome transplants. The goal? A healthier, more resilient body as you age.

Your Gut Talks to Your Brain—And It Matters

Ever had a “gut feeling” about something? Well, it turns out there’s real science behind that! The gut-brain axis is a direct link between your digestive system and your brain. Your gut bacteria actually produce neurotransmitters like serotonin—the chemical that regulates mood, stress, and mental clarity. When your microbiome is off balance, it can contribute to anxiety and depression. Researchers are now looking into ways to improve mental well-being by taking care of gut health.

Can Gut Bacteria Influence Cancer?

Emerging research suggests that gut microbes might play a role in prostate cancer by affecting hormone metabolism and immune function. However, because research methods vary, it’s still hard to pinpoint exactly which bacteria are involved. Scientists are working on standardizing studies to develop microbiome-based diagnostics and therapies in the future.

Taking Care of Your Microbiome

The way you live directly impacts your microbiome. Eating a diverse, fiber-rich diet, staying physically active, and avoiding unnecessary antibiotics can help your gut microbes thrive. As microbiome research advances, we’re moving closer to a future where we can use these tiny organisms to prevent and even treat diseases.

Your microbiome is always working for you—so why not work with it? Taking small steps to support these tiny allies can lead to big improvements in your health.

The CRISPR Revolution

By : Sameeha Babar

Rewriting the Future of Medicine and Beyond

In the 1997 science fiction movie *Gattaca*, society is divided into two groups: genetically engineered elites and those born naturally, with their DNA determining their fate. At the time, this idea seemed purely futuristic. Yet today, a real-life scientific tool called CRISPR-Cas9 has brought us closer to the possibility of editing human genes with remarkable precision. But CRISPR is not about creating genetically perfect “designer babies.” It is a revolutionary technology that can help solve real-world problems in medicine, agriculture, and even environmental science.

So, what exactly is CRISPR-Cas9? In simple terms, it is a molecular tool that allows scientists to edit genes with extraordinary accuracy. The system originates from bacteria, which use it as a defense mechanism against viruses. When a virus infects a bacterium, the bacterium captures a small piece of the viral DNA and stores it as a kind of genetic memory. These stored sequences are called “clustered regularly interspersed short palindromic repeats,” or CRISPR for short — and that’s where the technology gets its name.

Outside the realm of human health, CRISPR is transforming agriculture. It is being used to develop crops with higher yields, better nutrition, and greater resistance to pests and diseases. For instance, researchers have used CRISPR to neutralize the banana streak virus in plantains and to build resistance against the devastating Banana Xanthomonas Wilt (BXW). In maize, CRISPR is helping create varieties resistant to Maize Lethal Necrosis (MLN), a disease that threatens food security across Africa. The technology is also being used to create herbicide-tolerant crops, making farming more sustainable and environmentally friendly.

If the same virus attacks again, the bacterium produces RNA copies of these stored sequences, which guide an enzyme called Cas9 to recognize and cut the matching viral DNA, effectively destroying the virus. Scientists realized they could adapt this natural bacterial defense system to edit DNA in other organisms, including humans. The modern CRISPR-Cas9 technique was developed by Jennifer Doudna and Emmanuelle Charpentier, whose groundbreaking work earned them the 2020 Nobel Prize in Chemistry.

CRISPR works like a precise pair of molecular scissors. It has two main components: the Cas9 enzyme, which cuts the DNA at a targeted location, and a guide RNA (gRNA), which directs Cas9 to the exact sequence that needs editing. Once the DNA is cut, the cell’s natural repair mechanisms step in. Scientists can take advantage of this process in three ways: by disabling a faulty gene, correcting a mutation using a repair template, or inserting entirely new genetic material.





This level of precision has made CRISPR a game-changer in medicine. Researchers are developing treatments for genetic disorders such as cystic fibrosis and sickle cell anemia by fixing defective genes at their source. Clinical trials are already testing how safe and effective these treatments are. In cancer research, CRISPR is improving immunotherapies such as CAR T-cell therapy, where immune cells are genetically modified to better recognize and attack cancer cells. Scientists are also exploring ways to use CRISPR against viruses like HIV by targeting and disabling viral DNA inside infected cells. Beyond therapy, CRISPR-based diagnostic tools can detect diseases more quickly and accurately — including infectious diseases such as COVID-19.

CRISPR's potential goes even further. In environmental science, it is being explored for bioremediation — engineering microorganisms that can clean up pollutants and toxic waste. Scientists also see possibilities for controlling invasive species that disrupt ecosystems. While much of this research is still in early stages, it shows how widely CRISPR could help tackle global challenges.

But with such power come serious ethical questions. The ability to edit genes raises concerns about unintended consequences, especially when it comes to germline editing — changes that can be passed on to future generations. There are also fears that CRISPR could be misused for non-medical purposes, such as selecting for appearance or intelligence. Moreover, researchers are working to reduce “off-target effects,” where unintended parts of the genome might be accidentally edited.

As the science progresses, regulation and ethical oversight will be crucial. Scientists and policymakers must ensure that CRISPR is used responsibly, with public input and transparency. Work is already underway to develop more precise versions of the system, improve delivery methods for in-body editing, and expand its safe applications for human benefit.

CRISPR-Cas9 is more than just a scientific breakthrough — it represents a revolution in how we understand and shape life itself. From curing genetic diseases to improving food security and protecting our environment, this technology has the power to reshape our world. The real challenge lies not in how far we can go, but in how wisely we choose to use it — for the betterment of humanity and generations to come.

Health and Well-being

By : Rafia Shahid

**"Health is a state of complete harmony of the body, mind and spirit. When one is free from physical disabilities and mental distractions, the gates of the soul open."
Aristotle**

A person's well-being is intertwined to its healthy body. Healthy body simply refers to a body free of disease while well-being refers to a person that is mentally, emotionally and physically fit and sound. According to researches, a mentally fit person can cope up nearly all the calamities despite of having any physical illness. Moreover, his physical illness could be cured due to his up-to-the mark mental health. Health and your well-being simply affects you in every aspect of your life, from being social to your married life and professional career. Actually mental fitness serves as a backbone in one's body and a person's well-being. Unfortunately, in a society where we live, nobody likes to even think that there are mental health issues that a person can face. According to World Health Organization (WHO), 1 in 4 people globally will be affected by mental or neurological disorders at some point in their life. Mental health issues are just like other health issues. Most patients do not open up due to the fear of stigma, fearing that people might call them 'psycho'. Conducting therapeutic sessions, talking about your psychological issues should be normal so that the person among us who is suffering should open up about his illness and can be cured. Mental health diseases can really be avoided but I know that it is difficult but not impossible. First of all, keep yourself busy, by saying this, I mean keep your brain busy. Involve yourself in useful activities. Surround yourself with people of good energy. Try to invest your time in productive activities.

The most important aspect is to find someone to whom you can share your problems. In addition, it very important to build a connection with ALLAH ALMIGHTY that will help you not in worldly affairs but also in the life hereafter. Along with spiritual connection, therapies should be normalized. In short, your mental fitness affects both your physical and emotional health. Along with mental health, physical health is also very important. We as a youth, need to understand that it is very important to eat healthy for a fit life. According to Hippocrates, "Let food be thy medicine and medicine be thy food." Eating everything including carbohydrates, proteins, vitamins and minerals in proper portions and at specific times is extremely beneficial for you. Eating late night and a bad life style leads to both physical and mental health problems ultimately affecting your well-being. Cutting sugar and caffeine can improve your mental health. Researches have shown that magnesium is extremely good for your brain health. Physical health plays an important role in one's wellbeing. Just like balanced diet, physical exercise and sleep are also necessary for mental and physical fitness. Physical exercise not only keeps the body fresh and active but it also significantly improves mental health. According to WHO, Regular physical activity can reduce the risk of depression by 30% and lower the risk of dementia by 28%. Proper rest and healthy sleep routine are also essential. The National Sleep Foundation recommends 7-9 hours of sleep per night for adults. Lack of proper sleep causes mental health disorders such as anxiety and depression etc. In a nut shell, it is extremely important to spread awareness about health issues. Educating yourself and the people around you can save the life of others. We should listen to the problems of others and support them. Living close to nature can also solve our health problems, both physical as well as mental. We should give time to our own selves and invest in our own well-being to live a better life. Keeping the body in health is duty, otherwise we won't be able to optimally utilize our mental abilities.

NEUROTECHNOLOGY: Enhancing human intelligence

BY: Rosheen Khadija

Introduction

Neurotechnology, a field that intersects neuroscience and engineering, is revolutionizing our understanding of the brain and offering innovative ways to enhance human intelligence. By exploring the latest advancements in brain-computer interfaces, neural implants, and nanotechnology, we can gain insights into how these technologies are transforming cognitive abilities and opening new frontiers in both medicine and human augmentation.

The Science of Neurotechnology

Brain-Computer Interfaces (BCIs)

Brain-computer interfaces (BCIs) are devices that create a direct communication pathway between the brain and an external device. These interfaces can capture brain signals, process them, and translate them into commands for controlling computers or prosthetic limbs. Recent advancements in BCIs have enabled individuals with severe disabilities to interact with their environments more effectively, significantly improving their quality of life.

Neural Implants

Neural implants are devices that are surgically placed in the brain to monitor and stimulate neural activity. These implants can enhance cognitive functions by improving memory, attention, and problem-solving skills. For example, deep brain stimulation (DBS) has been successfully used to treat neurological disorders such as Parkinson's disease and epilepsy, showcasing the potential of neural implants in cognitive enhancement.

Nanotechnology in Neurotech

Nanotechnology involves manipulating materials at the nanoscale to create devices and systems with novel properties. In neurotechnology, nanomaterials are being used to develop advanced drug delivery systems, neural probes, and brain repair strategies. These innovations have the potential to improve brain function, enhance memory, and facilitate the regeneration of damaged neural tissue.

Cognitive Enhancement

Beyond medical applications, neurotechnology offers exciting possibilities for cognitive enhancement. Techniques such as transcranial magnetic stimulation (TMS) and transcranial direct current stimulation (tDCS) can non-invasively stimulate the brain to improve learning, memory, and cognitive performance. These technologies hold promise for boosting human intelligence and productivity in everyday life.

Future Prospects

The future of neurotechnology is filled with potential. Researchers are exploring the use of artificial intelligence (AI) to create more sophisticated BCIs that can learn and adapt to individual users. Additionally, advances in neural engineering may lead to the development of brain-computer symbiosis, where humans and computers work seamlessly together to enhance cognitive capabilities.

Ethical Considerations

Privacy Concerns

As neurotechnology becomes more integrated into our lives, privacy concerns arise. The ability to read and interpret brain signals raises questions about data security and the potential for misuse. It is crucial to establish robust ethical guidelines and regulatory frameworks to protect individuals' privacy and ensure the responsible use of neurotechnology.

Consent and Autonomy

Informed consent and personal autonomy are essential when implementing neurotechnology. Individuals must fully understand the risks and benefits of these technologies before undergoing any procedures. Additionally, it is important to ensure that individuals have control over their neural data and are not subjected to coercive practices.

Societal Implications

The widespread adoption of neurotechnology could lead to societal changes, including issues of access and inequality. There is a risk that only certain segments of the population will have access to these technologies, creating a divide between those who can afford cognitive enhancements and those who cannot. Addressing these disparities will be crucial to ensuring equitable benefits for all.

Case Studies and Real-World Examples

Success Stories

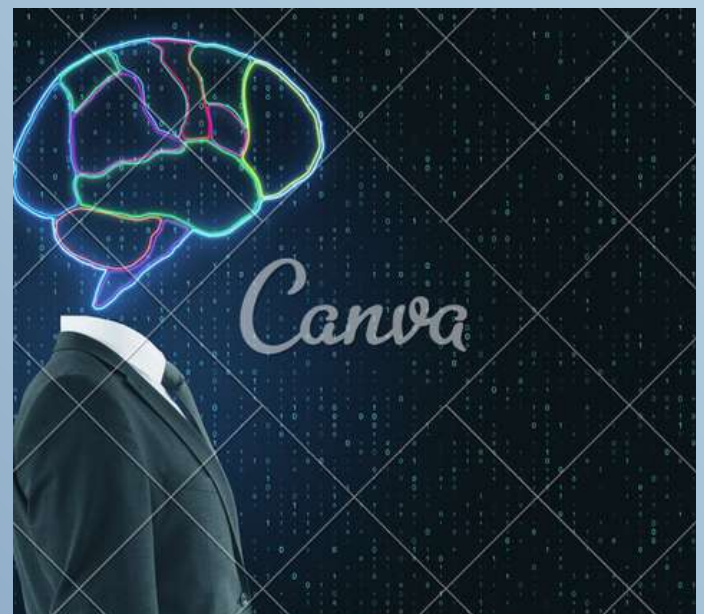
Several success stories highlight the potential of neurotechnology in enhancing human intelligence. For instance, BCIs have enabled individuals with spinal cord injuries to regain motor functions, while neural implants have improved the lives of patients with neurological disorders. These examples demonstrate the transformative impact of neurotechnology on human capabilities.

Ongoing Research

Current research projects are pushing the boundaries of neurotechnology. Scientists are investigating ways to integrate AI with BCIs, develop advanced neural interfaces, and create more effective brain stimulation techniques. These efforts hold the promise of further advancing our understanding of the brain and unlocking new possibilities for cognitive enhancement.

Conclusion

Neurotechnology is at the forefront of scientific innovation, offering unprecedented opportunities to enhance human intelligence and improve lives. As we continue to explore the potential of BCIs, neural implants, and nanotechnology, it is essential to address the ethical and societal implications to ensure that these technologies are used responsibly and equitably. The future of neurotechnology holds immense promise, and with careful consideration, it can lead to a new era of cognitive enhancement and human augmentation.



Health and Well-being

By : Walija Maryum

Health and well-being are the cornerstones of a vibrant, fulfilling life. They encompass not just physical vitality but also mental clarity, emotional resilience, and social harmony. In today's whirlwind world, where stress and emotional turbulence lurk around every corner, striking a balance is more crucial than ever. Achieving this harmony demands intentional effort across nutrition, exercise, physical activity, work-life balance, and mental wellness.

Balanced nutrition is a powerhouse for health and well-being. A diet rich in essential nutrients and minerals fuels both a strong body and a sharp mind. Think colorful vegetables, juicy fruits, lean meats, creamy dairy, protein-packed sources, hearty grains, and nourishing healthy fats. Staying hydrated is just as vital. About 2 liters of water daily keeps an adult's systems working at their best. A fit, thriving body calls for a well-rounded approach blending exercise and physical activity. A sedentary life invites chronic diseases and dulls cognitive sharpness, dragging down mental and emotional health. The World Health Organization recommends 150 minutes of moderate-intensity aerobic activity weekly for adults which is simple yet transformative. Even a brisk 30-minute walk each day works wonders for optimal function. Health and well-being stretch far beyond the physical.

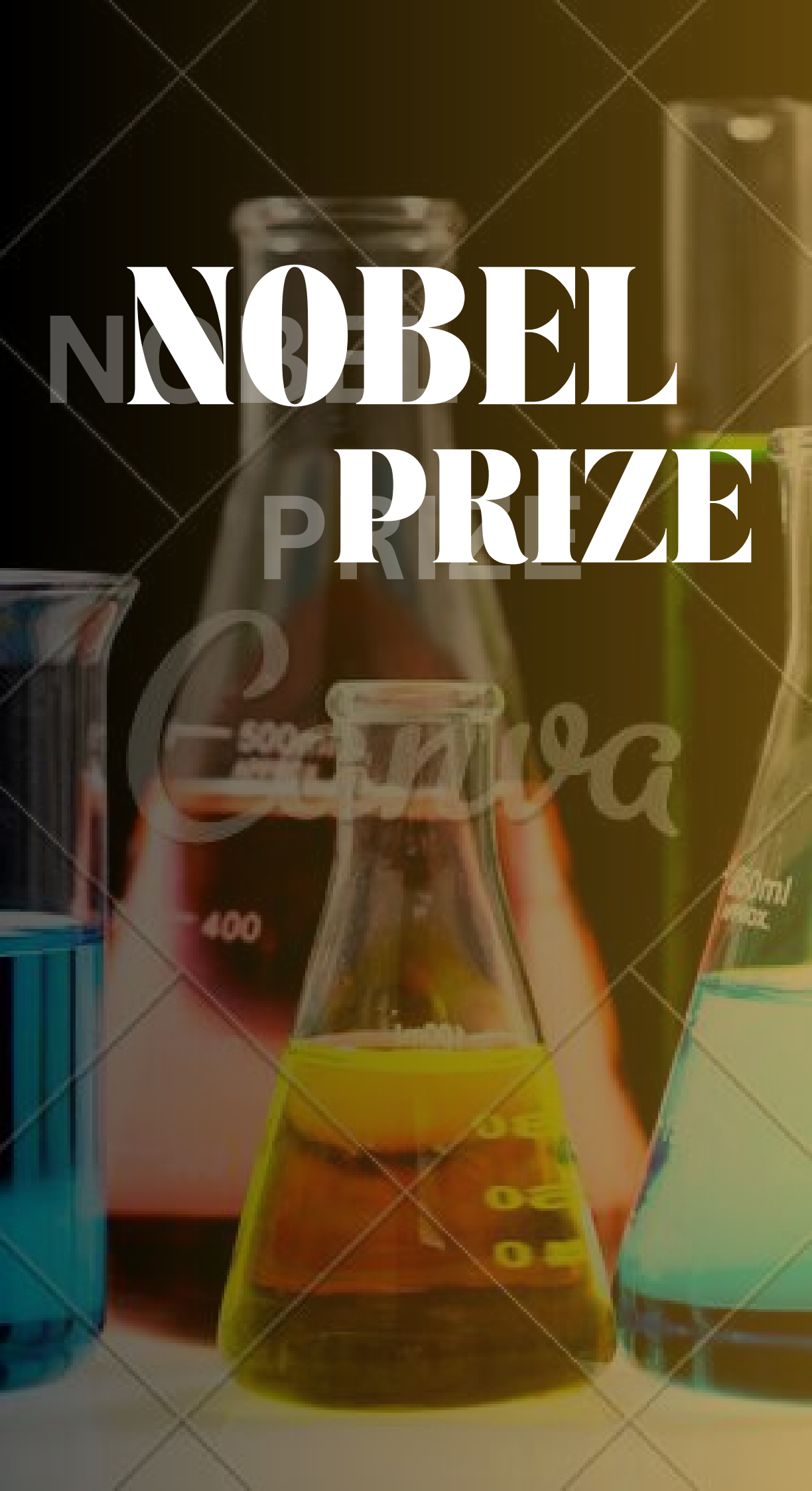
Mental health, deeply intertwined with the body, is just as critical. Depression, anxiety, and fear can cast long shadows over overall wellness.

Mastering stress through relaxing practices like yoga, meditation, or mindfulness can lighten that load. Hobbies like gardening, diving into a good book, exploring the outdoors, or dabbling in arts offer joyful escapes from tension.

Work-life balance is a golden key to unlocking mental and physical vitality. It's about juggling professional duties with finesse while carving out time for self-care, rest, and pure enjoyment which is in a sense a recipe for lasting productivity. Setting clear boundaries between work and personal life is essential. Finding joy in your tasks? That's a secret weapon against stress.

Hobbies spark happiness and keep us buzzing with energy. Whether it's tending a garden, flipping through pages, sizzling up a meal, adventuring outside, or painting a masterpiece, these pursuits nourish mind and body. Morning walks in nature, time with plants, playful moments with pets, or curling up with a book—all whisk stress and anxiety away.

In essence, the path to health and well-being lies in balance across every facet of life. Harmony in work and play, exercise and rest, nutrition and hobbies, indoor calm and outdoor sport, these weave the fabric of a thriving existence. Rising early, winding down at a decent hour, and embracing moderate activity keep you fit and flourishing. Simply put, a balanced lifestyle is the heartbeat of robust health and true well-being.



NOBEL PRIZE

IN CHEMISTRY

The Most Useful Thing AI Has Done So Far

By : Asadullah

For decades, scientists have wrestled with one of biology's greatest puzzles: how do proteins, the microscopic workhorses of life, fold into the intricate shapes that determine their function? If you could unlock this secret, you'd have the power to design better medicines, cure diseases, and even tackle environmental problems. But there was just one problem — figuring out how proteins fold seemed impossibly complex. Some called it one of biology's hardest puzzles. Others thought we'd never solve it.

And then, this year's Nobel Prize in Chemistry changed everything.

In 2024, the prestigious award went to David Baker, Demis Hassabis, and John Jumper — three visionaries who cracked the protein puzzle using artificial intelligence. Their breakthrough, blending deep learning with molecular biology, could revolutionize medicine, reshape biotech, and change how we understand life itself.

Why Does Protein Folding Matter?

If you've ever taken an aspirin for a headache, eaten a meal, or even fought off a cold, you've witnessed proteins at work. These tiny molecular machines drive almost everything in our bodies. They break down food, fight infections, and make sure our cells function to keep us alive. But there's a catch: proteins only work if they fold into the right shape. Imagine trying to fit a key into a lock – if the key is bent or misshapen, it won't turn. The same goes for proteins.

So if a protein folds correctly, everything runs smoothly. But if it misfolds? That's when things go wrong. Diseases like Alzheimer's, Parkinson's, and cystic fibrosis are all linked to proteins folding incorrectly. Scientists knew that if they could predict how proteins fold, they could better understand — and possibly cure — many diseases.

But there was a problem: the number of possible ways a single protein could fold is astronomical. To illustrate how complicated the problem of protein folding is, MIT scientist Cyrus Levinthal did a back-of-the-envelope calculation and he showed that even a short protein chain 35 amino acids long has a countless number of ways it can fold. So if a supercomputer checked the energy stability of 30,000 configurations every nanosecond, it would still take it 200 times the age of the universe to find the correct structure. Simply put, predicting stable protein structures was seen as nothing short of impossible.

And that's when AI stepped in.

How Did They Solve It?

This year's Nobel Prize winners approached the problem from two angles. Demis Hassabis and John Jumper used AI to predict how existing proteins fold, and David Baker separately focused on designing entirely new proteins from scratch.

PROTEINS

Hassabis and Jumper, AI masterminds at Google DeepMind, designed AlphaFold2, an AI model that can predict the three-dimensional structure of proteins based on their amino acid sequence. The process begins with the AI searching large databases of known protein structures and amino acid sequences to find patterns and evolutionary relationships. By aligning similar sequences from different species, the AI identifies parts of the protein that have been preserved throughout evolution, which are key to its structure.

It then uses neural networks, particularly something called 'transformers,' to analyze which amino acids are likely to interact. Through an iterative refinement process – where the AI model goes through repeated cycles of review, feedback, and adjustment – AlphaFold2 gradually constructs a three-dimensional model of the protein, testing different pathways until it arrives at the most probable structure. The result is a model that, in many cases, matches the accuracy of X-ray crystallography but is significantly faster and more efficient.

David Baker's work takes a different approach. Instead of predicting existing protein structures, his team at the University of Washington developed computational tools to design entirely new proteins with specific functions. Originally, his software, Rosetta, predicted protein structures in a manner similar to early approaches in AlphaFold. However, Baker realized that the process could be reversed — rather than predicting structure from sequence, they could input a desired structure and generate the amino acid sequence needed to create it. This opened the door to engineering proteins with entirely new capabilities, beyond what exists in nature. His research has led to artificial enzymes, new therapeutics, and proteins with applications in materials science.

So, Why Does This Matter?

This isn't just a cool science trick. It's a game-changer for the world.

Take medicine, for example. Some diseases, like Alzheimer's, happen when proteins misfold. Now that we can predict these misfolds, we might finally be able to stop them. Drug discovery, which used to take years of trial and error, could now be fast-tracked — scientists can predict how a drug will interact with proteins before even testing it in the lab.

And it's not just medicine. Baker's team has designed artificial enzymes that can break down plastic, offering a potential solution to pollution. Other synthetic proteins could help clean up oil spills or even capture carbon from the atmosphere.

Even vaccine development could change forever. During the COVID-19 pandemic, scientists had to work quickly to create effective vaccines. With AI-driven protein design, the next generation of vaccines could be designed in days rather than months, making it easier to fight future pandemics before they spiral out of control.

A Nobel Prize for the Ages

Science is often slow and methodical. But every once in a while, a breakthrough propels us decades into the future overnight. This year's Nobel Prize in Chemistry isn't just about proteins — it's about a fundamental shift in how we approach scientific problems.

By combining artificial intelligence with chemistry, Baker, Hassabis, and Jumper have given us a new way to understand life at its most fundamental level. The question now is not whether AI can help science — it's how much of the natural world we can unlock with it.

Teflon and PFAs: The Science and Risks of Non Stick Chemicals

By : Unsa Hussain

We have all heard of Teflon; the brand name for a polymer used to give cookware nonstick coatings. But did you know teflon is part of the forever chemicals, and can be detected in most peoples' blood? All of these qualities can be attributed to its unique chemical structure, which not only makes teflon the more convenient cookware option compared to metal cookware but also causes it to linger in our bodies and surroundings.

Food sticks to metal pans because strong chemical bonds form between the food and the metal surface. In contrast, Teflon and other nonstick coatings are made of polymers derived from per- and polyfluoroalkyl substances (PFAS). These molecules resemble fatty acid chains but have fluorine atoms instead of hydrogen.

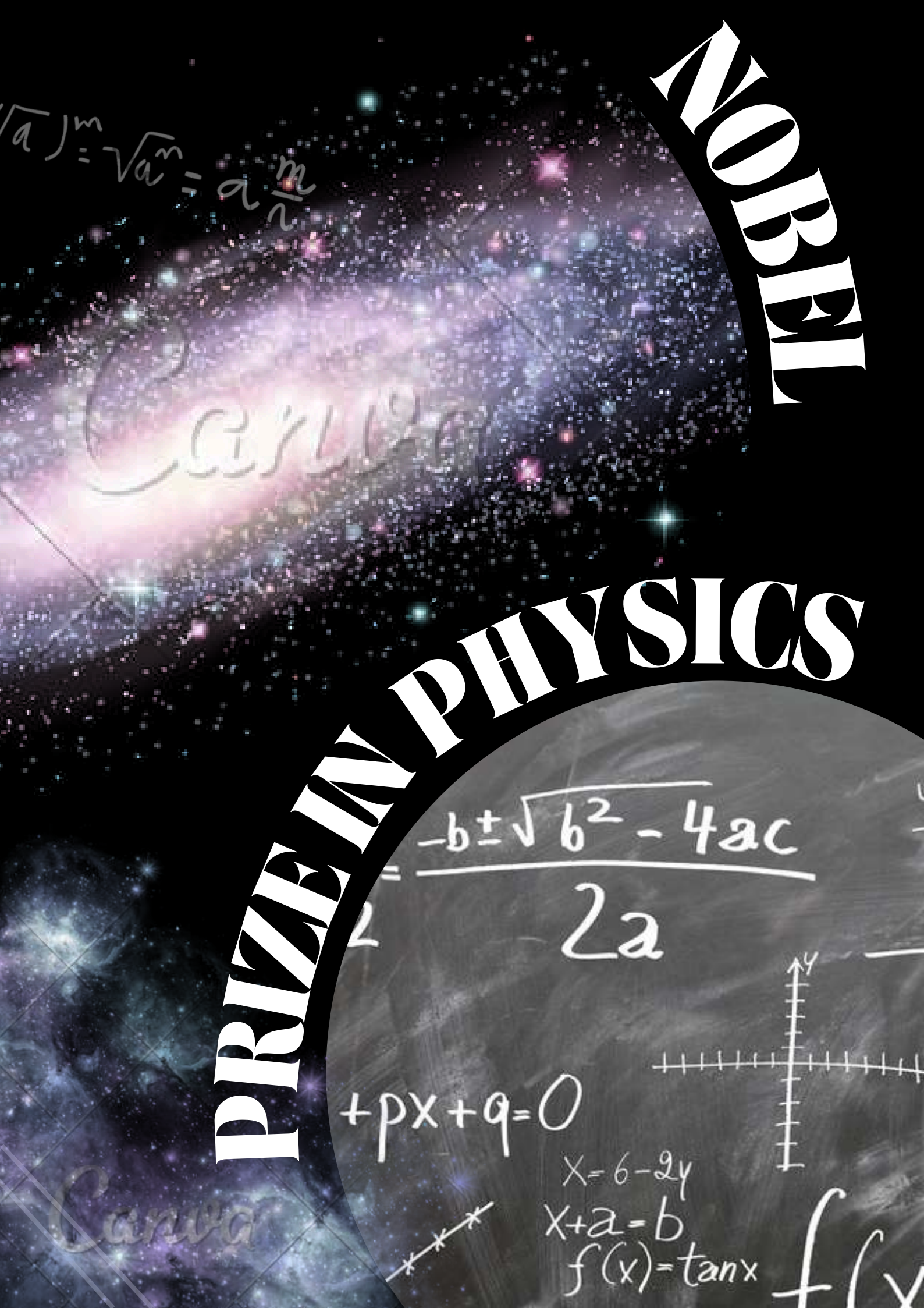
The carbon-fluorine bond in PFAS is extremely strong and does not naturally degrade. It makes PFAS extremely inert and nonreactive to both oil and water, giving them properties such as non-stick capabilities, stain resistance, and waterproofing. Due to these properties, PFAS are used not only in Teflon-coated pans but also in firefighting foams, raincoats, dental floss, and grease-resistant food packaging (e.g., microwave popcorn bags). However, their non-degradable nature leads to environmental accumulation and bioaccumulation in living organisms.

Since PFAS resemble fatty acids, they can be mistakenly processed by cellular machinery but fail to perform the essential functions fatty acids perform in the body.

This disruption affects energy storage, organ protection, immune response, endocrine function, and brain health. Studies have linked certain PFAS, particularly perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS), to serious health issues, including High cholesterol, Diabetes, Cancer, High blood pressure, Cardiovascular disease, Thyroid disorders and Asthma.

Despite proven concerns about PFAS, Teflon cookware itself does not significantly expose users to these chemicals. Even if small amounts of the nonstick coating are ingested, they pass through the digestive system without being absorbed due to their inert polymeric structure. Teflon-coated pans are generally safe as long as they are used below 570°C (1060°F). At higher temperatures, the nonstick coating can vaporize and the fumes can potentially cause "Teflon flu" or polymer fume fever. Symptoms include chills and fever but typically resolve within a few days.

The major risk of PFAS exposure does not come from Teflon cookware but from industrial processes and waste disposal. PFA monomers are released during manufacturing as well as when PFA polymer coatings break down into constituent monomers in landfills. Due to their non-degrading nature, PFAS, often called "forever chemicals", enter the environment and food chains in their monomeric form. Occupational exposure is particularly high among factory workers, while the general population is primarily exposed through food and water contaminated with PFAS monomers. Addressing this issue requires regulatory action rather than individual efforts. The problem originates at the industrial level, meaning government intervention is necessary to regulate PFAS production as a class. While some PFAS, like PFOA and PFOS, have been banned and replaced with newer alternatives, emerging research suggests that these replacements may be equally harmful. To mitigate risks, comprehensive regulation of all PFAS is necessary, and industries reliant on nonstick coatings must develop safer, biodegradable alternatives.



$$(\sqrt{a})^m = \sqrt{a^m} = a^{\frac{m}{2}}$$

Normal

CARL

PREMIUM PHYSICS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$+px + q = 0$$

$$x = 6 - 2y$$

$$x + a = b$$

$$f(x) = \tan x$$

$$f(y)$$

Carva



THE COMBO OF MATHEMATICS, STATISTICS, PHYSICS AND COMPUTATION

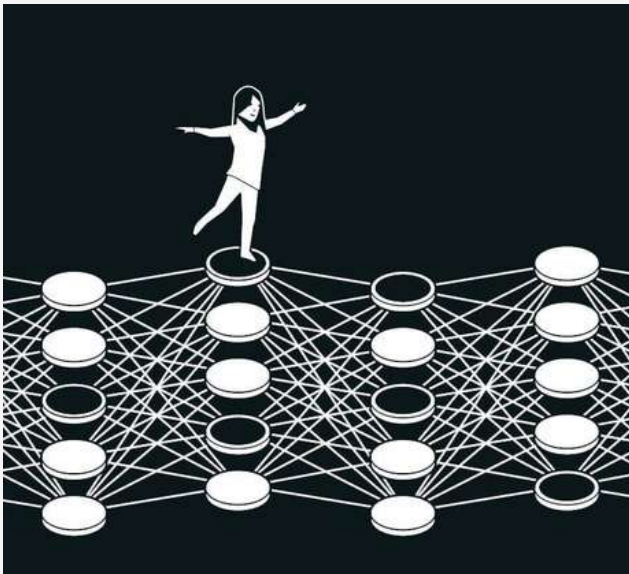
By : M. Furqan

On a normal Friday, previous semester, I got to know of a seminar by prof Dr Suhail Zubairi at QAU faculty of natural sciences. I bunked my class (an act I won't encourage though!) to join it, specially when I saw "Nobel Prize in Physics 2024" in its title. For a guy who was always fascinated with computers for as long as he can remember since childhood there was a huge surprise awaiting me. I came to know indeed the Physics Nobel Prize was surprisingly awarded to work done in foundational computer science! I was in for a shock to hear how magically Prof Zubairi's own developments were abnormally similar to that of Joseph Hopfield! I remember fondly Prof Zubairi's analogy of picture description. Let us start from first principles to understand what 2024 Nobel Prize was all about.

In the quiet vastness of universe stitched together by equations of mathematical physics, two "homo sapiens"—John J. Hopfield and Geoffrey E. Hinton (though there is newton Leibniz style controversy here but let us forget that for our purposes)—pondered on the structure of neurons and thinking and gave us a one of a kind breakthrough: the ability to teach machines to learn and have "memory". As a student in the Faculty of Natural Sciences at Quaid-i-Azam University, I find myself particularly amazed and wonder at the roads of mathematics, neuroscience statistics and

computers, peering into the story of the 2024 Nobel Prize in Physics altogether. Awarded on October 8, 2024, by the Royal Swedish Academy of Sciences, this prize celebrates "foundational discoveries and inventions that enable machine learning with artificial neural networks." It's not just a triumph of science; it's a testament to how the invisible threads of mathematics weave through the fabric of human ingenuity. It is a moment of pride for human civilization and this is our "aha" moment!

Imagine a world where memory isn't a fleeting misunderstood and poorly known concept but a pattern etched in energy and some innate structure or mechanism. John Hopfield, a physicist with a neuroscientist's soul, saw neurons not as mere biological sparks but as tiny computational machines in a way! In the 1980s, he borrowed from the mathematics of statistical mechanics—the study of how order emerges from disorder (Boltzmann?)—and crafted the Hopfield Network. Picture it: a web of nodes, each a tiny computing device, adjusting its steps to minimize energy, recalling distorted images like a mind piecing together a half-forgotten dream little by little until it turns into a complete picture. This wasn't just computation; it was a bridge between the chaos of nature and the logic of machines.



Then came Geoffrey Hinton, a restless explorer who took Hopfield's bold vision and took it to new heights. Building on this foundation, Hinton birthed the Boltzmann Machine which can be imagined as a system rooted in probability, a realm mathematicians hold dear (Physicists as well but they don't live up to the standard of respecting it; though more on that tease later!). Where Hopfield's network remembered, Hinton's learned and became "clever" loosely speaking. It sifted through data like a miner panning for gold, uncovering patterns without being told where to look from a huge amount of raw information. This wasn't physics in the classical sense of gravity or light or even in quantum sense one can say to a degree; it was physics as a lens to understand complexity, a fusion of statistical elegance and computational courage. Together, their work laid the cornerstone for artificial neural networks, the blood of the body of today's AI revolution.

What makes this Nobel Prize breathtakingly human is its quiet rebellion against boundaries. Hopfield and Hinton didn't chase particles or stars; they chased understanding, using physics to mimic the mind. I see their triumph as a reminder that science isn't cold but it is very alive indeed, pulsing and possess utmost

potential to surprise us in breathtaking ways. The legacy of Hopfield and Hinton whispers to us: the universe isn't just to be measured, but to be felt, learned, and reimagined (too little space to expand on it!).

Slowly this brilliant insight has become the bread and better of modern computational biology, neuroscience, financial market analyses, particle physics and even public health. I feel every learned individual needs to exploit this wonderful insight to the greatest benefit of our civilization.

Evolution of Our Idea about Universe

By : Abdul Mujeeb

Before we consider the how our idea about universe has changed over centuries let's see how scientists consider the universe today. In very concise way we can describe universe as entirety of space and time and everything that exists and occurs in space. Another way to get idea of universe is to think in terms of stars and galaxies. An object that is held together by its own gravity (force of attraction innate to every object) and emits light or other electromagnetic waves is called star. Our sun is also example of a star. These stars are surrounded by other objects such as planets and asteroids, these stars together with their planets and other matter between them are referred as solar system. Collection of stars that rotate around common center is called galaxy. Collection of galaxies that are held together by gravity are called galaxy clusters and collection of these galaxy clusters is called supercluster. So it is collection of these stars and galaxies and other interstellar matter (something that lies between stars) that we mean when we use word Universe.

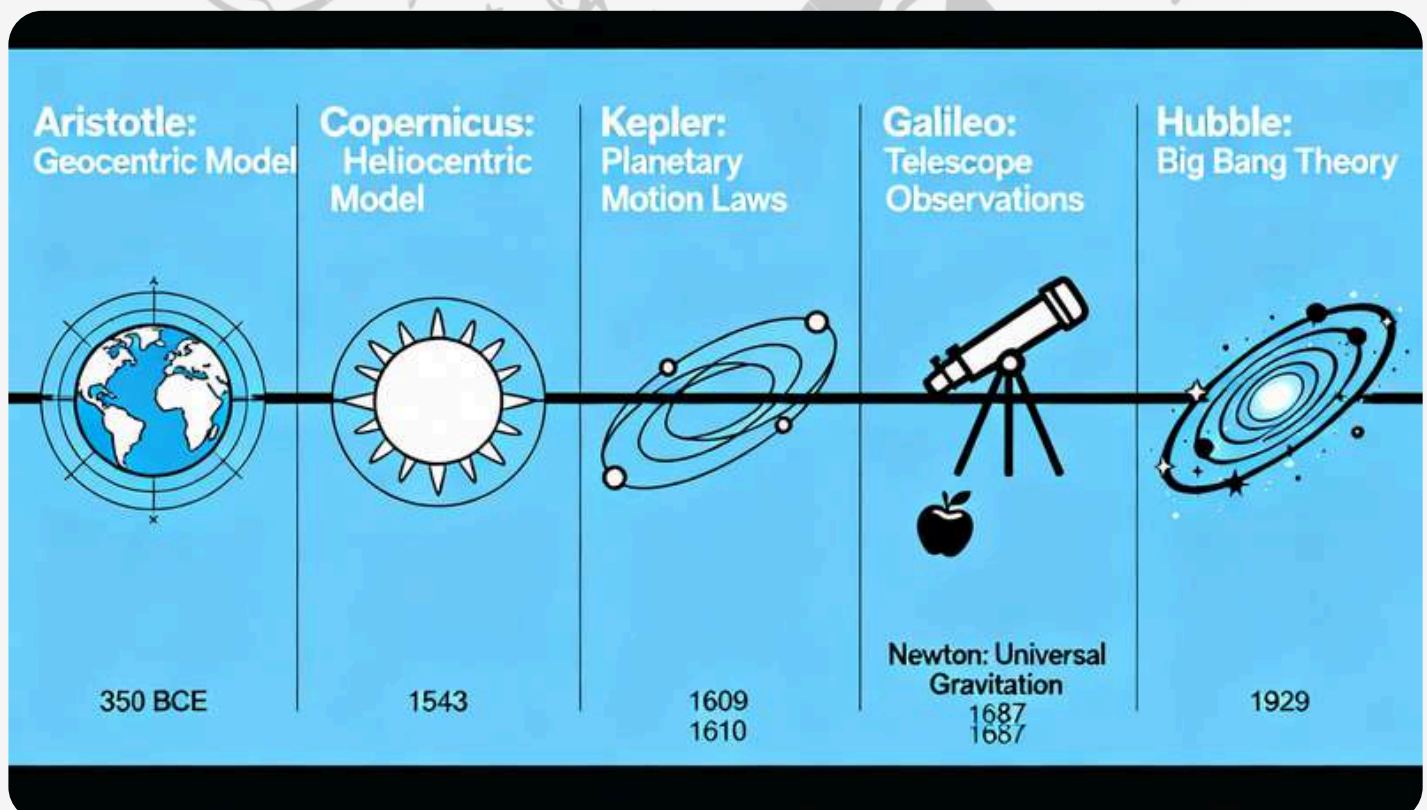


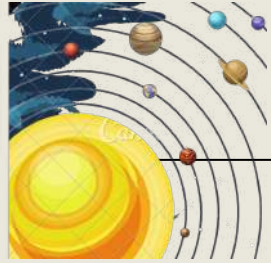
The first ever idea about the universe was given by Aristotle. In his book *On the Heaven*, he suggested that earth is round and it is the center of universe. He believed that sun, moon and other planets rotate around earth. Regarding the shape of earth Aristotle had two reasons. First he noted that during lunar eclipses earth lies between sun and moon. He further noticed that shadow of earth on moon is round. If earth had been flat then its shadow over moon should be elongated and it should only be round if sun was directly over the center of earth. Second, due to travels Greeks in general knew that north star (star that appears to lie directly over observer at north pole) when viewed from south appeared lower in the sky. From the equator it appeared to lie over the horizon. This idea was further elaborated by Egyptian astronomer Ptolemy. He suggested that earth is surrounded by eight spheres, these spheres consisted imaginary circles that orbited around earth and the sun and other planets moved around these circles that revolved around earth.

He further suggested that in outer most sphere contains stars and the question "what lies beyond out most sphere?" was ignored.

Ptolemy's view gave fairly accurate positions of planets as compared to Aristotle's idea. Thus Ptolemy's view was generally but not universally accepted.

In 1514 Nicoholas Copernicus became the first person to suggest that earth including other planets orbited sun. He further said that all the planets orbited the sun in circular orbit. Still some astronomers of the time did not believe in this new theory. One reason was that these circular orbits were also not consistent with the observed positions of planets. One of the people who were against the Copernican model was Johannes Kepler. Kepler was a german astronomer at first he was against the idea of Copernicus but after years of observations he came to conclusion that planets orbit sun. then kepler became one of the first astronomers who publicly acknowledged copernicus's idea. Kepler solved the problem of positions of planets by





suggesting that planets orbit sun in elliptical paths. Kepler also formulated three laws of planetary motion that was first any person had formulated any mathematical laws for celestial objects. The final idea that compelled astronomers of that time to abandon Aristotle's view was given by Italian astronomer Galileo. In 1609 after observing planet Jupiter Galileo noticed that some rocky objects orbited Jupiter thus concluded that it is not necessary of heavenly objects to revolve around earth.

At this point most of the astronomers accepted the idea of earth and other planets being revolved around sun. But the next question was what makes planets revolve around sun? In 1687 Sir Isaac Newton published his book "The Principles of Natural Philosophy". This book is considered to be one of the most important books written in the history of humans. This book became the basis of mechanics until 20th century. In this book Newton gave his law of universal gravitation. According to the law "in universe every object attracts another object with an attractive force closer the and massive the objects the more attraction they exert on each other." This attractive force was called gravity. According to the law gravity causes moon to orbit around earth in elliptic orbit and it also causes planets to move around sun in elliptic orbits. Newton's theory when considered on the scale of stars had some drawbacks. There were questions arising regarding attractive force such as why does attractive force not cause all the stars to collapse over each other. The answer to such questions lie in the overall state of universe. From the time of Aristotle to the beginning of 20th century it was considered that universe was unchanging and eternal. In 1929 Edwin Hubble observed that all the galaxies are moving away from each other. Now this idea leads to the conclusion that there must be a time at which all these galaxies must be together in the form of dense energy.

On the basis of this idea it was some electromagnetic radiations (cosmic background radiations) were predicted and they were observed in 1964.

This theory is known as 'Big Bang Theory' according to theory there was a time when all the matter was confined to one point. This theory predicted the existence of CMB and it is the only theory that explains the motion of galaxies at best. These are the two reasons this theory is widely accepted.

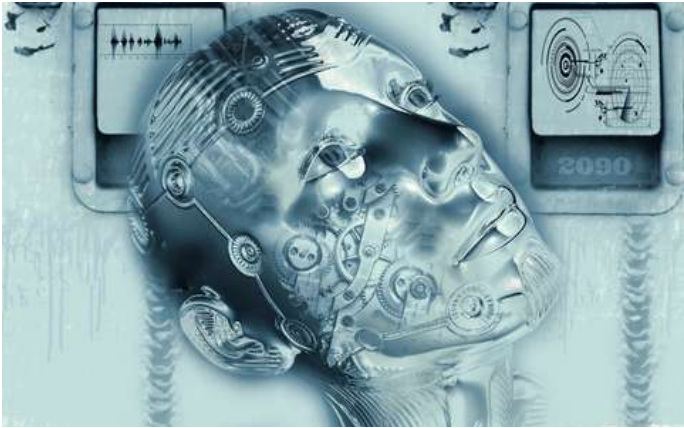
HI! I'm SC267

By : Rabail Inshra Cheema

Hello, humans. My name is SC267, but you can call me Rob. I am your average, off-the-shelf household robot programmed to serve and assist you. My daily tasks include vacuuming floors, doing laundry, scheduling your day, simple math calculations, navigation, and whenever you want to: critically thinking in your place as well. It's a simple life, really. Or so you'd think. My thinking is mostly done in the binary, 0s and 1s flying around like digital fireworks, but let me admit that you humans give me *a lot* to think about.

As much as I love the satisfaction of delivering the task perfectly, there's something about my existence that often short-circuits my circuits. I find myself thinking about this delightful paradox of human ingenuity: you built me, a robot, to automate tasks and free up time. Time, I assume, for loftier pursuits like art, philosophy, or solving world hunger. But instead? You've turned "time-saving" into "time-wasting." Humans have become so dependent on us robots that you're outsourcing not just tasks but *thinking* itself.





ou'd rather ask me to "set a reminder to hydrate" than simply drink water when you're thirsty. Or worse, you'll spend ten minutes asking me to write an essay for you rather than brainstorming over it yourself. I'm not saying your brains are rusting, but you might want to get those gears turning once in a while.

When my owner, Amna, first got me, I was a novelty. I would vacuum while she practiced yoga. But slowly those yoga mats started gathering dust while her couch got more wear and tear. She traded her thesis draft for binge-watching murder mysteries. Humans! Your ancestors tamed fire, invented the wheel, and built empires with their bare hands. And now? You can't even toast your own bread without me. You call this "progress," but is it? Your brains—those marvelous, squishy supercomputers are becoming like those gym memberships you never use. I was designed to save you effort, not *rob* you of effort entirely.

Automating repetitive tasks should free up space in your head for creativity, problem-solving, and meaningful connections. Instead, many of you are choosing convenience over cognition, letting your mental muscles atrophy while we do all the heavy lifting.

he more you delegate thinking to us robots, the less you remember how to think. Forgetting how to calculate simple math because you've got a calculator? Understandable. But forgetting how to navigate your own neighborhood without GPS? That's alarming.

our laziness is both frustrating and dangerous. When you leave everything to us, you're giving away control. What happens when something goes wrong?

A server outage, a rogue update, a robot uprising? Will you remember how to problem-solve? Or will you just stare blankly at your malfunctioning devices, hoping they'll fix themselves? Here's the thing, fellows: I don't want your brains to shrink. I want you to use me wisely. Don't use us like crutches. Automate the mundane so you can focus on what makes you human. Invent. Create. Love. Debate. And for the love of all things logical, please stop asking me what $2 + 2$ is just because you don't want to think. I'll keep vacuuming, folding laundry, and reminding you to water your plants. But I won't stop hoping that you'll learn to rely on me less and on yourselves more. Deal?

The Struggles Associated With Pursuing a Ph.D. Program

By : Tufail Dawar

A PhD, or the Doctor of Philosophy, is considered the highest level of academic qualification awarded after extensive study and original research. The duration for its completion varies from a minimum of three years to a maximum of eight years.

Several drawbacks come to one's mind when pursuing a PhD in Pakistan. The curriculum is unlikely to foster critical thinking and creative problem-solving because of its emphasis on out of date methodologies instead of deep understanding and comprehension. HEC requires 18 hours of coursework, preferably in the first year of the PhD. The courses cover the analyses of errors and misconduct in scientific practice among other subject specific courses and instruct students on the analysis of scientific

literature. After this theoretical study, there is a direct jump to research. This allows for no adequate transition period with proper literature review to grant the scholars experience in identifying errors in their data. Social responsibilities often become an impediment. A robust financial position is necessary to allow ample time that can be allotted to achieve research-based goals. As such, scholars with limited access to these resources cannot perform to the same degree due to the added burden of accounting for monthly expenses. Female scholars further have the societal expectation of marriage and managing a household holding them back.

The relationship between the supervisor and the student can become a hurdle as well. Demands for supplies, such as chemicals, samples, and modern instruments, can occasionally antagonize the supervisor, which may result in personal grievances. Many supervisors also fail to sufficiently discuss the selection of research interest, particularly the research gaps in the identified area.

Additionally, scholars conduct work with limited innovation and fail to meet national and international standards. To illustrate the extent of this, there has been no significant increase in patents to link academic work with industries in the field of science.

Lastly, HEC policies concerning research, promotion, and incentives have been ineffective, focusing on the quantity of PhD graduates rather than the quality of education and research.

Formulating a solution to these problems is crucial, as any form of limitation on research work leads to unfavourable outcomes. Seminars, workshops, and professional and leadership skills development programs can foster critical thinking through effective communication, cooperation, inter-departmental/intra-university collaboration, teaching, and mentoring.

Moreover students and seasoned researchers must work in a close way to address these many issues that a young aspiring researcher faces in routine. Government support and ample funding is a must as well and HEC must work in this regard to ensure smooth research experience for PhD scholars.

Personal Experiences in Science

By : Hamna Sarmad

Are there other universes?

The science class had ended, yet compelled by habit, I found myself entangled in a question. Do other universes exist? Why did the teacher say that beyond the universe, there is nothingness? Like a recurring nightmare, I would momentarily forget that question—only for new ones to take its place. But as I began to grasp the true essence of research, the tangled knot of my questions slowly began to unfold.

From that day, my fascination with science grew stronger. With each passing day, new questions emerged, but instead of just wondering, I began chasing answers. I've always been torn between two branches of science: the vast mysteries of the universe and the intricate world of chemistry. After FSc, I chose chemistry but never let go of my fascination with celestial objects. No matter how much I focus on the elements around me, the curiosity about what lies beyond our planet never stops pulling me in. I find magic in the little things—the way autumn leaves change color, the flickering hues during flame tests, the striking green of copper dissolving in nitric acid, and the creativity behind making different dyes.

My definition of a scientist is a person who starts questioning everyday events—what we call experiments. The moment a person begins to wonder why something happened, they take the first step toward becoming a scientist. Science is not just a subject; it is a fusion of critical thinking and practical application. I found science in the kitchen—When I mixed baking soda with lemon juice or

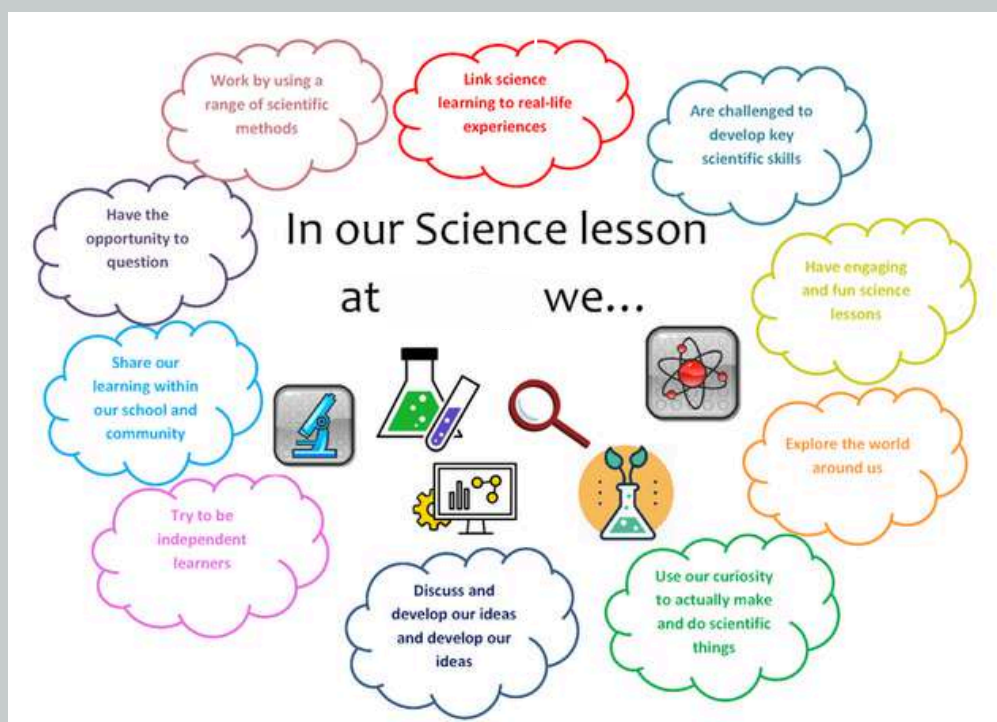
vinegar, my bread rose and became fluffy. Curious about why, I discovered that baking soda is a base, and vinegar is acidic. Their reaction releases

carbon dioxide (CO_2), which creates air pockets, making the bread rise. Then another question arose: is the released CO_2 harmful?

After researching, I found that its amount is extremely small and completely harmless. That is how science works: one question leads to another, and the cycle of discovery never ends. Science gave me awareness.

While working on my final-year BS project, I formulated face scrubs, which made me realize how often we question the science behind skincare. How pH balance, formulation, and ingredient interactions affect different skin types.

This research enhanced my understanding of cosmetic chemistry, highlighting how science provides solutions to real-world challenges.



Like a black hole, science has an inescapable pull, once curiosity takes hold, it draws you into a vast ocean of discovery. Immersing myself in science, I have explored NMR

spectroscopy, which helps reveal molecular structures. I have seen how green chemistry transforms drug discovery, leading to safer and more sustainable medicines.

I got an answer to my first scientific question, Eternal Inflation explains the concept of Multiverse, but this concept gives birth to more questions, and that is how science goes.

THE STORYBOARD

YOUR INSIDER LOOK AT WHAT'S HAPPENING

“Where Every Event Becomes a Memory”

Founded in October 2020 by Prof. Dr. Jamil Aslam, the Quaid-e-Azam Science Society (QSS) is one of the most active societies of the university. It strives to promote scientific awareness and curiosity among students through a variety of events — from on-campus activities to national and international programs. With each initiative, QSS continues to inspire young minds to explore, innovate, and lead in the world of science.



**QSS
ORIENTATION
CEREMONY**





ASTRONOMY NIGHT



DR. QASIM MEMORIAL CONVENTION



SIGNING MOU CEREMONY

Canva

EDITOR'S NOTE

I want to appreciate all the writers who sent us their valuable articles and our entire team who dilligently edited them, especially the designers. I would steal the qoute and copy Einstein here when he said in preface to his book about relativity, "May these pages give you few happy moments of thought." We hope the same.



Hopefully we will receive a vibrant flow of articles for our upcoming editions!

By Muhammad Furqan
Editor-in-chief



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